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The role of executive functions in 9- to 12-year-old children's sentence processing: An eye-movement study

Nannan Cui

School of Psychology, Northeast Normal University, Changchun, China

Yang Wang

School of Psychology, Northeast Normal University, Changchun, China and School of Psychology, Xinjiang Normal University, Ürümqi, China

Jiefei Luo

School of Psychology, Northeast Normal University, Changchun, China

Yan Wu

School of Psychology, Northeast Normal University, Changchun, China

Background: Executive function (EF) plays a crucial role in children's reading. However, previous studies were based on offline products of reading comprehension. Online research is needed to reveal the core mechanisms underlying children's reading processing. By measuring children's working memory (WM) and cognitive flexibility (CF), we investigated whether individual differences in EF could modulate sentence processing and, if so, how they exert their roles.
Methods: The present study manipulated semantic congruency and the association between crucial words in a sentence. We recruited 89 Chinese children aged 9–12 years and monitored their eye movement.
Results: The study revealed distinct associations between reader- and text-related characteristics, as evidenced by eye-movement patterns during reading. A significant incongruency effect was observed in reading, underscoring the children's capacity to discern incongruent information. Children's WM and CF were found to modulate this process. Specifically, high-WM children showed more effective integration of incongruent information when the textual context was closely related during the later-stage processing. In contrast, low-WM children faced more challenges with incongruent words. Additionally, CF was influential during the early processing period. High-CF children exhibited longer early-stage reading times for incongruent words in associated contexts.
Conclusions: Individual differences in EF can modulate children's online sentence processing. However, different EF components may play different roles.

Keywords: executive functions, eye movement, sentence congruency, word association

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Highlights

What is already known about this topic

- A positive association exists between children's EF and reading, which often relies on offline products of reading comprehension.
- In children's sentence processing, sentence congruency and word association are the text features that require specific attention.
- The few studies investigating whether individual differences in EF modulate the processing of sentence congruency and word association have almost exclusively involved adults.

What this paper adds

- Children exhibit longer reading times for incongruent words, highlighting the cognitive demand of incongruencies.
- WM plays a modulatory role during the later stages of sentence processing.
- CF influences children's reading during the early processing period.

Implications for theory, policy or practice

- EF is a crucial cognitive basis that supports children's reading processing.
- WM and CF can underlie learning differences, especially when the focus of reading instruction switches from 'learning to read' to 'reading to learn'.
- Reading interventions that support reading-related WM and CF may be especially beneficial for children.

Reading is a complex, goal-directed activity involving the simultaneous orchestration and integration of various processes (Cartwright et al., 2020). The factors influencing reading include the reader, text, activity and sociocultural context (Kieffer et al., 2013). Recent studies have focused on the role of reader characteristics, such as executive function (EF), in reading comprehension (for a review, see Follmer, 2018). However, these studies have primarily employed offline questionnaires to examine the relationship between EF and children's reading ability, which may reflect the results of cognitive processes only during the reading that elicited the responses. In addition, most extant studies that investigated the role of EF on reading comprehension by measuring readers' cognitive processing were conducted with adults (Boudewyn, 2013; Kaakinen et al., 2002; Van Petten et al., 1997; Yang et al., 2020). Moreover, online research regarding children's reading development has focused predominantly on word and text characteristics (Rayner, 1998). Therefore, the present study examined both reader and text characteristics and investigated whether and, if so, how EF modulates children's sentence processing. This was measured using the eye-movement technique, which continuously tracks online cognitive processes during reading.

EF refers to a set of top-down control processes associated with the prefrontal cortex that facilitate planning, problem-solving and the initiation and maintenance of goal-directed

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behaviour (Diamond, 2013). EF is generally assumed to include three related yet distinct higher order cognitive dimensions: working memory (WM), inhibitory control and cognitive flexibility (CF) (Miyake et al., 2000). Some researchers have proposed that individuals' EF, as a domain-general cognitive ability, is one of the core factors that modulates children's reading processes. For instance, the cognitive view of reading (CVR; Kendeou et al., 2014) asserts that reading comprehension involves lower and higher level processes. Low-level processing refers to the acquisition of meaningful language units from visual symbols. In contrast, high-level processing combines different language units into meaningful and coherent mental representations that rely on attention allocation, reasoning and EF (Zargar et al., 2020). The two-component model of evaluative comprehension further describes when and how EF plays a role. The model defines the reading process as consisting of two stages: an initial comprehension stage (e.g., evaluating accuracy and detecting difficulty) and a later evaluative stage (e.g., attempting to correct errors and resolve the difficulty) (van Moort et al., 2021). The later repair process, which may be constrained primarily by individuals' WM and other cognitive control skills, is typically slow, resource demanding and reader initiated (Richter, 2015).

Online reading processing theories predict a crucial role for EF in individuals' reading (e.g., Schroeder et al., 2008). However, few empirical studies have explored this issue. Specifically, eye-movement technology has been extensively utilised to investigate the online reading process. Eye-movement monitoring offers insights into the moment-to-moment comprehension of linguistic material in a natural reading context (Rayner, 1998, 2009). Various parameters serve as indicators of underlying processes, aiding in differentiating between early detection and later evaluation stages. For example, the first fixation duration (FFD) on a word in the first encounter or the amount of time spent on a word with one or multiple fixations before leaving it for the first time (gaze duration [GD]) is thought to capture the initial decoding process and lexical access or incongruency detection (Connor et al., 2015; Zargar et al., 2020). Conversely, second reading time (SRT) represents the cumulative GD on an area after its first reading, with zero value if not revisited (Clifton et al., 2007). Regression probability measures the likelihood of a reader regressing from a target word and is binomially coded. Both serve as indicators of in-depth semantic coherence integration or assessment (de Leeuw et al., 2016a, 2016b).

Prior eye-movement studies on reading have primarily concentrated on word and text characteristics, such as word length, frequency and predictability (Kliegl et al., 2006; Rayner, 1998). Although individual differences (IDs) in basic processes underlying language are ubiquitous, current theories and experimental methods often overwhelmingly downplay their importance (e.g., by relegating them to error variance) (Kidd et al., 2018, p. 154). Cognitive skills contributing to IDs include language-specific factors, such as vocabulary knowledge, phonological awareness and word reading, and general cognitive factors, such as WM and reasoning skills (de Leeuw et al., 2016a, 2016b; Kim et al., 2020; Kuperman et al., 2018; Veldre & Andrews, 2015). However, whether and how IDs in the general cognitive system influence language processing is particularly intriguing, as it illuminates the impact of children's extralinguistic cognitive system on language processing.

With the eye-movement technique, de Leeuw et al. (2016a, 2016b) explored the role of text and student characteristics in real-time reading processes for children aged 7 years 8 months to 12 years. In these two studies, WM, the ability to hold information while performing one or more mental operations, was measured as a student characteristic. During the experiments, de Leeuw et al. examined either the average reading time for each word

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(de Leeuw et al., 2016a) or the eye movement of gaze, look back and SRT for the heading, first sentence and final sentence in a passage (de Leeuw et al., 2016b). However, they found mixed results regarding the effects of WM on eye-movement behaviours. In de Leeuw et al.'s (2016a) study, high WM resulted in longer GD for third graders and shorter GD for fifth graders. However, in their second study (de Leeuw et al., 2016b), which included a backward digit span task to measure WM, no significant correlation was found between WM and either GD or SRT. One explanation for the mixed results may be the confounding effects of short-term memory (STM; measured mainly by a forward digit span task in these two studies), as a high correlation exists between WM and STM tasks, and the authors considered the predicting effects of the two factors at the same time. In fact, when measuring children's listening span only, Kim et al. (2020) found that WM capacity was negatively related to GD and SRT among third graders, indicating that students with high WM showed shorter reading times.

Based on this evidence, we know that WM may be associated with children's online sentence processing. However, these findings did not reveal the exact role of WM or other EF components during the reading process. Some empirical findings involving adults showed that WM was not only associated with individuals' reading time but also modulated the integration of sentence congruency and semantic association processing (Kaakinen et al., 2002; Van Petten et al., 1997; Yang et al., 2020). Sentence congruency refers to word meanings woven into the context to form a cohesive message (Camblin et al., 2007). For example, Yang et al. (2020) found that readers with low WM showed a more negligible incongruency effect than participants with high WM, suggesting that low-WM participants were fully occupied when processing the current word and did not notice semantic incongruence between existing words and prior contexts.

Moreover, by manipulating whether the text information is relevant to the readers' perspectives, Kaakinen et al. (2002) found that low-span readers showed a perspective-relevant effect in both GD and SRT, regardless of the material's familiarity. In contrast, high-span readers showed a perspective-relevant effect only in GD and only for non-familiar material. Further analyses revealed that, for low-span readers, the perspective effect was due to faster GD of irrelevant information and increased look-back time on relevant information. For high-span readers, the perspective effect for non-familiar material resulted from increased GD of relevant information in the text. The findings support the idea that high-span readers more effectively allocate their attentional resources to relevant information than low-span readers. During initial reading processing, high-span readers devoted more effort to relevant information, whereas low-span readers did so later on (Kaakinen et al., 2002).

Additionally, after manipulating whether a current word is semantically associated with its preceding words by sharing a common schema, Boudewyn (2013) observed that individuals with high WM were less sensitive to word association and thus could integrate meaning quickly without relying on detailed information, such as word association. However, this study was conducted on listening comprehension. Interestingly, when word association and sentence congruency were manipulated within the same experiment, Van Petten et al. (1997) found that WM would modulate the integration of sentence congruency but not word association processing. When participants were first separated into three groups (i.e., high, medium and low span) according to their scores on a reading span task, Van Petten et al. (1997) determined that all groups showed significant effects for word association. In contrast, the effects of anomalous sentences were observed only in the high- and medium-span groups. The authors argued that associative relationships between single

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words developed from repeated co-occurrences over time and thus did not require the formation of new conceptual combinations in WM. In other words, the effect of WM on relevant information processing remains inconclusive.

Nevertheless, existing evidence regarding the modulation effects of EF on sentence congruency and relevance processing focuses only on adult participants. As children differ from adults in various ways, including the reading process and cognitive skills (Schroeder et al., 2015), eye-movement research comparing adults' and children's reading processes reveals that children typically have more and longer fixations and engage in rereading more often than adults (for a review, see Schroeder et al., 2015). Thus, conclusions from studies on adults cannot be generalised to children. Moreover, most extant studies focused only on WM. However, EF also includes inhibitory control and CF (Miyake et al., 2000). Relative to WM and inhibition, CF occurs much later in child development (Garon et al., 2008). CF generally refers to the ability to flexibly shift attention between mental sets, operations and tasks (Diamond, 2013). Reading processing requires CF, which helps the reader shift text representations and eventually develop a new substructure of the mental model (Kieffer et al., 2013).

Of the three core EF factors, WM is the most well studied, but CF is perhaps the most important component affecting children's reading comprehension development (Kieffer et al., 2013). As an EF subcomponent distinct from WM (Miyake et al., 2000), CF may play a unique role in reading. Comparing the predictive effects of WM and CF on children's reading comprehension, prior research has concluded that CF plays a potentially stronger role (e.g., Kieffer et al., 2013). However, such evidence was derived mainly from offline measurements of reading comprehension. Presently, how WM and CF modulate the reading processes and whether CF plays a more dominant role than WM in children's reading processing are uncertain.

Thus, the present study investigated the influence of WM and CF on sentence reading processes, including the timing and mechanisms of their effects. Specifically, our participants were 9- to 12-year-olds transitioning from word decoding to text comprehension (Helder et al., 2016), making how children read at this stage a major concern of our research. Moreover, our study measured nonverbal intelligence as a control variable because previous research has demonstrated its positive relationship to reading comprehension in children (Naglieri, 2001).

In the present study, we manipulated text-relevant variables, that is, sentence congruency and word association. Sentence congruency is when word meanings integrate into a context, creating a cohesive message. For example, the sentence 'Mother stuck the envelope on the stamp for later use' presents an incongruency. Comparing reading times for congruent versus incongruent statements often reveals an incongruency effect: Readers typically take more time processing incongruent sentences than congruent ones. Word association usually refers to the semantic relationship between preceding words and current words, as they share a common schema. In the previous example, 'stamp' and 'envelope' represent an associated condition, whereas 'stamp' and 'calendar' create an unassociated condition (Camblin et al., 2007). Specifically, our objectives are (1) to determine if sentence congruency and word association influence children's real-time sentence processing; (2) to assess whether IDs in WM or CF modulate the processing of congruency and association; and (3), if modulation occurs, to pinpoint at which stage (i.e., early or late) WM or CF exerts its influence.

Based on Camblin et al.'s (2007) and Zargar et al.'s (2020) findings, we hypothesised that sentence congruency would impact children's real-time sentence processing, with

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word association serving as a moderator. Specifically, we expect both the main effect of congruency and its interaction with word association to be significant. Second, we predicted that WM would modulate the processing of sentence congruency and word association, showing the significant two-way or three-way interactions among WM, sentence congruency and word association. However, the interactions may only be significant during the reprocessing period (reflected by the SRT or regression probability), as predicted by the two-component model (Richter, 2015). Moreover, given that CF is the foundation for EF (Diamond, 2013) and helps readers shift text representations and develop new mental models (Kieffer et al., 2013), we hypothesised that CF would also modulate sentence congruency and word association. Namely, the interaction between CF and sentence congruency, between CF and word association or the three-way interaction among these three factors would be significant for reading times or regression probability.

Methods

Participants

Drawing from the studies of Camblin et al. (2007) and de Leeuw et al. (2016a, 2016b), we employed G*Power software for our statistical power calculations (Faul et al., 2009). The results showed that when the test power was 95% and the reference effect size was $f = 0.25$, $\alpha = .05$, the sample size was 52 participants. From a public school in China, 89 students (44 girls), with a mean age of 10.39 years (range: 9–12 years), participated in this study. All were native Chinese speakers with normal or corrected hearing and vision, no language impairment or other neurological disorders. Written consent was obtained from each child's parent or guardian before the experiment. The Institutional Review Board of the Northeast Normal University granted ethics approval for this study.

Measures

Eye-Movement Study: Experimental Design and Materials

A factorial design crossed two levels of word association and sentence congruency, resulting in four conditions (see Table 1). In the congruent-associated condition, the two critical words (e.g., stamp and envelope) were embedded in a meaningful sentence. The incongruent-associated condition was formed by reversing the order of the two critical words. In the two unassociated conditions, the two critical words (e.g., stamp and calendar) were semantically unrelated. Similarly, the incongruent-unassociated condition was constructed by reversing the order of the two unrelated words.

For the word pairs in the four conditions, we first chose simple nouns from Chinese textbooks for primary school students and recruited 30 students to judge the familiarity of these words using a 3-point Likert scale (1 = *know the word and its meaning* to 3 = *do not know the word*). The results showed that these words would be familiar to participants ($M = 1.05$, $SD = 0.09$). Finally, 48 group words were selected for the experiment, with each group containing three types of words: the core word, an associated word pair and an unassociated word pair. As shown in Table 2, three types of critical words were matched for stimulus properties (Fs < 2.32, ps $> .10$).

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Table 1. Examples of experimental materials and plausibility rating results.

Sentence type	Example	Rationality rating
Congruent associated	妈妈把*邮票*贴在信封上以备下次使用。	3.20 (0.35)
	Mother stuck the *stamp* on the envelope for later use.	
Incongruent associated	妈妈把*信封*贴在邮票上以备下次使用。	1.60 (0.38)
	Mother stuck the *envelope* on the stamp for later use.	
Congruent unassociated	妈妈把*邮票*贴在挂历上以备下次使用。	3.07 (0.45)
	Mother stuck the *stamp* on the calendar for later use.	
Incongruent unassociated	妈妈把*挂历*贴在邮票上以备下次使用。	1.58 (0.33)
	Mother stuck the *calendar* on the stamp for later use.	

Note: The values in parentheses are the standard deviations.

Table 2. Stimulus example and properties of three type target words.

	Core word	Associated word	Unassociated word	*F*/*t*	*p*
Example	邮票 Envelope	信封 Stamp	挂历 Calendar		
Mean stroke number	17.63 (3.96)	16.08 (4.27)	16.04 (4.09)	2.32	.10
Mean log character frequency	4.59 (0.73)	4.81 (0.57)	4.69 (0.71)	1.03	.36
Mean log word frequency	3.13 (0.71)	2.91 (0.70)	3.02 (0.80)	1.06	.35
Familiarity	1.08 (0.31)	1.05 (0.24)	1.03 (0.18)	2.10	.13
Semantic relatedness	–	3.32 (1.05)	2.14 (1.26)	15.21	**<.001**

Note: The values in parentheses are the standard deviations. Bloded *p*-values signify statistically significant results. The word frequency library was obtained from the Center for Chinese Linguistics at PKU.

Second, 25 additional students were recruited to rate the semantic relatedness of the word pairs (1 = *completely unrelated* to 4 = *completely related*). As expected, the semantic relatedness between associated word pairs was significantly higher than between unassociated word pairs ($t[47] = 15.21$, $p < .001$, Cohen's $d = 2.10$). Next, 39 additional children were recruited to rate the degree of semantic acceptability across conditions on a 1 to 4 scale (1 = *something improbable in the real world* to 4 = *something highly probable in the real world*). As shown in Table 1, there was a significant main effect of congruency ($F[1, 47] = 814.90$, $p < .001$, $\eta_p^2 = 0.95$), with the rating level in the congruent condition significantly higher than in the incongruent condition.

Finally, word pairs were used to construct 48 groups of experimental sentences. To gauge the predictability of target words (the second critical noun highlighted in Table 1), we asked 23 third- to fifth-grade children to complete sentences with the first word that came to their mind. The cloze probability denotes the percentage of participants who completed a given sentence with a specific word (Taylor, 1953). The congruent-associated and congruent-unassociated groups averaged cloze probability values of 6.61% and 2.16%, respectively. Target words for incongruent sentences were not reported. Additionally, another 17 children were recruited to complete the predictability assessment for the first critical nouns, resulting in a consistent predictability rate of 0 across all conditions, indicating very low predictability for all conditions (Block & Baldwin, 2010).

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Following a Latin square design, we divided experimental sentences into four lists, with each containing 48 items (12 sentences per condition). To conceal the purpose of the experiment from participants, each list included 40 filler sentences whose structures matched the experimental sentences. Comprehension questions (20 total) followed some of the sentences to determine whether participants were attentively reading for comprehension or inattentively skimming.

Apparatus and Data Analysis. An Eye Link 1000 Plus tracker was used to record eye-movement data and present sentences in 28-font white, regular scripts on a black background. Utilising SR Research Data Viewer, we meticulously processed the eye-movement data. Key steps included (1) removing extremely short (<60 milliseconds) or long (>800 milliseconds) fixations as per the Clean Data function (Cui et al., 2013), excluding 7.3% of data; (2) eradicating interruptions such as blinks, tracker losses and sentences with <5 fixations, further omitting 1.2%; and (3) discarding observations beyond 3 standard deviations, which resulted in an additional 3.9% data removal, ensuring robust analysis quality. In total, the analysis included 4272 trials. Our primary focus was on the effects of the target words (the second critical noun highlighted in Table 1).

Executive Functions

Working Memory. WM was measured using a 2-back memory task (Owen et al., 2005) with a digit from 1 to 9 randomly presented on the screen. Beginning with the third digit, participants were asked whether the current digit matched the digit before the previous one. Participants pressed a 'yes' key for a match (target digit) and a 'no' key for a mismatch (non-target digit). After the number was presented for 2 seconds or the students pressed a key, the subsequent trial was conducted. The 2-back task consisted of 200 trials, including 20 practice trials. After the practice session, the subjects began the formal experiment, which consisted of two blocks. The items' accuracy rates were recorded for further analysis. The reliability of the assessment was adequate (α = .823).

Cognitive Flexibility. Children's CF was measured using an odd, even, more than or less task (Zhao et al., 2018). Single digits between 1 and 9 (except for 5) were centrally presented on a computer monitor, with students instructed to indicate whether the number was odd or even (2, 4, 6 and 8 for even; 1, 3, 7 and 9 for odd) when the number of words was presented in green. Additionally, they were asked to indicate whether the digit was more or less than five when the number of words was presented in red. Thus, the trials required students to shift between these two categorisation operations. For all trials, the students had to press the choice button with speed and accuracy using different hands, according to one of the two modelled tasks. The number of correct responses measured the task outcomes. The reliability of the assessment was adequate (α = .871).

Nonverbal Intelligence

We used Standard Progressive Matrices (Raven et al., 1998) to measure children's nonverbal intelligence. The raw score was the number of correctly answered items. The reliability was adequate (α = .769).

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Procedure

All assessments were conducted in quiet locations at participants' schools over three testing periods (two individual and one group). After attending the eye-movement experiment, all children completed two EF tasks, followed by an intelligence test (group). The entire testing session lasted approximately 1.3 hours per child.

Statistical Analysis Strategy

Due to the fact that students and sentences are crossed, each data point represents a unique combination of subject and item. Linear mixed-effects models (LMMs) were used to analyse random intercepts and slopes for both subjects and sentences (Baayen et al., 2008).

Model construction and analysis were performed using the lme4 package in R (Bates et al., 2014). We employed LMMs to analyse the durational measures, specifically FFD, GD and SRT, as they represent continuous reading time data. Conversely, due to its categorical nature, regression probability was analysed using a logit LMM (de Leeuw et al., 2016b). Log-transformed values of reading time measures (FFD, GD and SRT) were used as dependent variables, with separate models fitted for each index. To address the first research question regarding the impact of congruency and association without factoring in IDs, we constructed models with sentence congruency and word association as fixed factors. Deviation contrast coding (−0.5 and 0.5) was used for the factors of categorical variables (i.e., sentence congruency and word association).

Subsequently, we incorporated individual variables to assess the moderating roles of WM and CF. In order to investigate the interactions between WM/CF and text-relevant variables while controlling for the effects of other variables, all variables were entered as predictors within the one model. This enabled stronger conclusions to be drawn about the relative contribution of WM and CF. The fixed factors in this model included two within-subjects variables (i.e., sentence congruency and word association) and four between-subjects variables (i.e., age, intelligence, WM and CF).

The full model was constructed according to the principle of maximal random effects (Barr et al., 2013). If the full model failed to converge, the random correlation term was removed. If convergence was still not achieved, principal component analysis was used to compare the variance of the random effects in the model, remove random effects with contributions equal to or less than zero and simplify the model until convergence was achieved and no odd-fit warnings were reported (Brown, 2021). Table 5 presents the fixed effects of the best-fit models. For a summary of the random-effects structures, refer to Appendix A.

Results

The average comprehension score for the remaining children was 84% (ranging from 60% to 100%). All scores were significantly higher than 50% (the chance level, ts > 17.00, ps $< .001$), suggesting that all children were genuinely comprehending and not merely skimming the sentences. The descriptive results of all predictors are presented in Table 3.

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Table 3. Summary of intercorrelations, means and standard deviations for predictor variables.

Variables	1	2	3	4	*M*	*SD*
1. Age	–				10.39	1.03
2. Intelligence	.22*	–			0.72	0.12
3. Working memory	.12	.12	–		0.83	0.11
4. Cognitive flexibility	.09	.25*	.26*	–	0.78	0.13

*$p < .05$.

Table 4. Means and standard deviations (in brackets) of semantic congruency and word association.

	First fixation duration	Gaze duration	Second reading time	Regression probability
Congruent associated	230 (86)	381 (218)	245 (324)	0.17 (0.37)
Incongruent associated	236 (91)	409 (232)	297 (358)	0.21 (0.41)
Congruent unassociated	226 (88)	389 (232)	255 (328)	0.17 (0.38)
Incongruent unassociated	229 (87)	404 (234)	316 (353)	0.22 (0.42)

Effects of Sentence Congruency and Word Association

Table 4 presents the means and standard deviations of reading time as a function of sentence congruency and word association. Before incorporating ID variables, LMM analyses revealed a significant main effect of sentence congruency on the GD, SRT and regression probability of the target word region. Reading times were longer in the incongruent condition than in the congruent condition ($ts > 2.67$, $ps < .008$). Additionally, the incongruent condition had a higher regression probability ($z = 3.05$, $p = .002$). After accounting for ID variables, the effect of congruency remained significant (see Table 5), suggesting a very stable effect of word congruency even after controlling for children's age, intelligence and other factors. However, in both models, the main effect of word association and its interaction with congruency were not significant.

Regulation Effects of WM

For the regulation effects of WM, the only significant interaction among WM, sentence congruency and word association occurred for SRT ($\chi^2 = 5.38$, $df = 1$, $p = .020$). The total number of SRT trials in LMM was 2194, demonstrating that students reread critical words in more than half of the cases (51.36%). Simple effects analyses showed that for children with high WM ($n = 44$, $M = 0.92$, $SD = 0.05$), the SRT of incongruent target words was significantly longer than that of congruent target words in the unassociated condition ($\beta = -0.07$, $SE = 0.03$, $t = -2.26$, $p = .027$) but not in the associated condition ($\beta = -0.01$, $SE = 0.03$, $t = -0.38$, $p = .704$). By contrast, children with low WM ($n = 45$, $M = 0.75$, $SD = 0.08$) spent more time rereading the incongruent words in the

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Table 5. Fixed effects in the linear mixed-effects models for eye-movement measures.

	First fixation duration				Gaze duration				Second reading time				Regression probability			
	β	*SE*	*t*	*p*	*β*	*SE*	*t*	*p*	*β*	*SE*	*t*	*p*	*β*	*SE*	*t*	*p*
Intercept	2.33	0.01	393.12	<.001	2.53	0.01	248.22	<.001	2.58	0.01	233.45	<.001	−1.31	0.10	−12.99	<.001
Age	0.00	0.01	0.58	.57	−0.02	0.01	−1.73	.09	−0.03	0.01	−2.63	.01	0.12	0.09	1.37	.17
Intelligence	−0.05	0.06	−0.86	.40	−0.09	0.12	−0.74	.46	−0.04	0.09	−0.42	.68	−0.07	0.77	−0.09	.93
Working memory (WM)	0.03	0.05	0.47	.64	−0.19	0.12	−1.56	.13	0.12	0.10	1.10	.27	−0.10	0.83	−0.12	.90
Cognitive flexibility (CF)	−0.03	0.05	−0.59	.56	0.09	0.08	1.05	.31	−0.06	0.09	−0.64	.52	0.97	0.71	1.38	.17
Sentence congruency	0.01	0.01	1.69	.09	0.02	0.01	3.33	**<.001**	0.05	0.01	3.95	**<.001**	0.24	0.08	3.03	**.002**
Word association	−0.01	0.01	−1.60	.11	0.00	0.01	0.06	.95	0.00	0.01	0.33	.74	−0.01	0.08	−0.10	.92
WM:congruency	0.01	0.05	0.24	.81	0.11	0.07	1.43	.16	−0.04	0.13	−0.30	.77	0.10	0.80	0.13	.90
WM:association	−0.03	0.05	−0.49	.63	−0.09	0.08	−1.06	.30	−0.15	0.16	−0.97	.36	0.56	0.76	0.75	.46
CF:congruency	0.07	0.05	1.52	.14	0.05	0.06	0.71	.48	−0.07	0.11	−0.59	.56	0.33	0.68	0.48	.63
CF:association	0.02	0.05	0.40	.69	−0.04	0.07	−0.52	.61	−0.09	0.12	−0.70	.49	0.05	0.65	0.08	.93
Congruency: association	0.00	0.01	0.09	.93	−0.01	0.02	−0.52	.61	−0.01	0.03	−0.25	.80	0.01	0.18	0.03	.98
WM:congruency: association	−0.06	0.11	−0.58	.56	−0.22	0.17	−1.31	.19	0.79	0.34	2.32	**.03**	−0.88	1.68	−0.52	.60
CF:congruency: association	−0.19	0.11	−1.72	.10	−0.35	0.14	−2.43	**.02**	−0.07	0.26	−0.25	.81	−1.20	1.50	−0.80	.43

Note: *p*-values that are bolded indicate statistically significant results.

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associated condition ($\beta = -0.09$, $SE = 0.03$, $t = -3.24$, $p = .002$) compared with the unassociated condition ($\beta = -0.02$, $SE = 0.03$, $t = -0.68$, $p = .501$), as shown in Figure 1.

This finding indicates that associated textual content aids children with high WM in integrating incongruency information during subsequent readings, eliminating differences between congruency and incongruency conditions. Conversely, for children with low WM, the more associated the content, the more challenging the integration.

Regulation Effects of CF

As shown in Table 5, the significant interaction among CF, sentence congruency and word association was only shown in GD ($\chi^2 = 5.92$, $df = 1$, $p = .015$). Children with high CF ($n = 44$, $M = 0.88$, $SD = 0.04$) took more time to read incongruent words in the associated condition ($\beta = -0.06$, $SE = 0.01$, $t = -4.34$, $p < .001$) compared with the unassociated condition ($\beta = -0.02$, $SE = 0.01$, $t = -1.14$, $p = .259$). However, no significant difference occurred for the low-CF ($n = 45$, $M = 0.69$, $SD = 0.11$) group for either the associated ($\beta = 0.004$, $SE = 0.02$, $t = 0.19$, $p = .850$) or unassociated condition ($\beta = -0.02$, $SE = 0.02$, $t = -1.17$, $p = .246$), as depicted in Figure 2.

These findings suggest that children with high CF can quickly detect incongruencies during early processing, compared with those with low CF.

Power Analyses

Post hoc power analyses for our study design were performed using the simr package (Green & MacLeod, 2016). The results indicated that at a significance level of .05, the statistical power to detect the three-way interaction was .65 for GD involving CF, sentence congruency and word association and .81 for SRT involving WM, sentence congruency and word association.

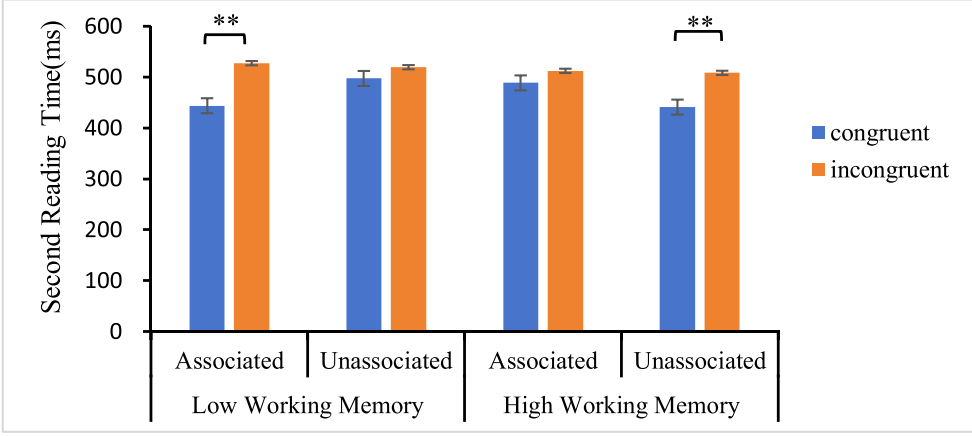


Figure 1. Regulation effect of working memory on second reading time. *Note*: **$p < .01$.

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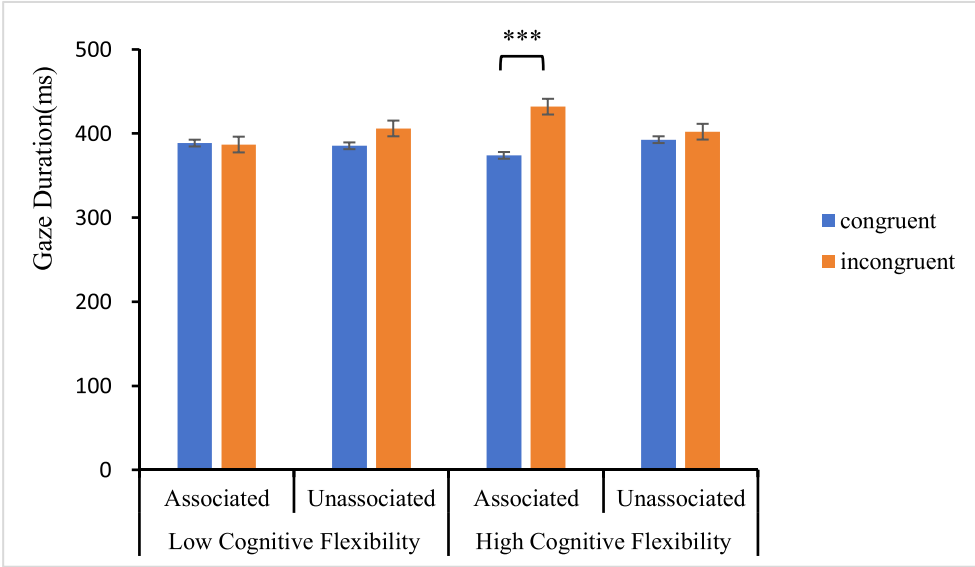


Figure 2. Regulation effect of cognitive flexibility on gaze duration. *Note*: ***$p < .001$.

Discussion

By recording children's eye movements, the present study explored whether children's online sentence processing was associated with their EF. Furthermore, the study examined whether sentence congruency and word association processing varied in children with different WM or CF. Results revealed a clear incongruency effect, with children displaying extended reading times and a higher regression probability for incongruent versus congruent target words. During the later stages of sentence processing, as evidenced by the SRT metric, WM appeared to play a modulatory role. Associated text information facilitated the processing of incongruent words for high-WM children but not for low-WM children, who faced greater challenges. Additionally, CF played a role during the early processing period. High-CF children displayed longer early-stage (GD) reading times for incongruent words, especially in the associated contexts.

Effects of Sentence Congruency and Word Association

The complexity of sentence comprehension in children's reading becomes clear when we consider sentence congruency and word association. Sentence congruency significantly affects children's reading. Incongruent target words consistently yield longer reading times across metrics, such as GD and SRT, and also exhibit a higher regression probability. Our findings align with previous research on children's self-monitoring during comprehension (Connor et al., 2015). However, word association does not show a significant impact. One reason for this may be the significant differences among children in WM or CF. Indeed, we found significant regulating effects of WM/CF on the processing of sentence congruency and word association. Another reason may be that children prioritise sentence congruency over word association. When confronted with both, their attention gravitates towards congruency (Van Petten et al., 1997).

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Regulation Effects of WM

We explored whether, when and how WM modulates the processing of sentence congruency and word association. After controlling for age, intelligence and CF, our LMM analysis revealed no main effect of WM. This differs from Kim et al. (2020), who noted faster reading among students with higher WM compared with their lower WM peers. The discrepancy may stem from each study's focus. Kim et al. (2020) emphasised average reading time across the whole text without defining a specific region of interest, while our research focused on the reading time of a target word, which may not be sensitive enough to represent children's reading efficiency.

However, we found that individuals with different WM abilities exhibit differences in the integration of textual information. High textual content relevance can facilitate the integration of incongruent information by children with high WM, but it can also cause a processing burden for those with low WM. This effect is evidenced by high-WM children showing no significant incongruent effects during word association, while low-WM children show more significant incongruent effects. We speculate that semantic association information enabled high-WM children to rapidly integrate incongruous words, streamlining the reading process and diminishing the need for deep analysis of incongruency. Conversely, semantic association information in context sentences imposes a processing burden on low-WM children's reading comprehension due to their limited WM capacity, which makes it difficult for them to integrate incongruent information.

In fact, these findings align with the CVR view, which highlights the pivotal role of WM for understanding essential details during sentence comprehension (Kendeou et al., 2014). The capacity theory of comprehension (Just & Carpenter, 1992) posits that efficient information manipulation is key for sentence comprehension due to the constraints of WM resources. Therefore, individuals with high WM can process multiple sources of information, whereas those with low WM may focus only on dominant types of semantic information. Our results align with this argument. Children with high WM were able to process two types of semantic information (i.e., sentence congruency and word association). However, individuals with low WM focus only on sentence congruency, which is assumed to be the dominant semantic information in a sentence (Van Petten et al., 1997).

Regulation Effects of CF

After controlling for age, intelligence and WM, we found no significant main effect of CF on reading times, though we did observe that CF could regulate the processing of incongruent information. In the earlier word processing stage, as reflected by GD, children with high CF quickly detected the incongruent words and integrated them into sentences, especially when the semantic content of the text was closely related. In contrast, children with low CF could not immediately detect such incongruent information.

As CF is the ability to flexibly switch between and among different tasks, operations or cognitive sets (Miyake et al., 2000), researchers have suggested that it is particularly critical for integrating information from multiple sources during sentence reading (Follmer, 2018). Moreover, CF enables readers to use rereading and information-searching strategies to support sentence comprehension (Follmer, 2018; Kieffer et al., 2013). Our study also suggested that children with high-CF skills were more proficient at using semantic information in sentence processing. Additionally, we found that CF can play a regulatory role in the early detection stages of sentence processing. This differs from the two-component model, which

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posits that EF plays a role in the late evaluation stage of sentence processing (Richter, 2015). Our findings suggest that EF is a necessary component of the reading process and plays a role in whole-sentence processing.

In fact, our findings indicated that, compared with WM, CF may play a more significant role in sentence processing. As our results show, WM played a role at the late stage of sentence processing, whereas CF exerted regulating effects during the early processing period. This implies that CF may enable readers to identify and resolve problems earlier than WM does. Notably, this finding is in line with the results of some offline tests (Follmer, 2018; Knudsen et al., 2018). For example, Kieffer et al. (2013) found that CF contributes to children's reading comprehension more than other EFs. In most previous sentence processing studies, WM has received relatively more attention than CF (de Leeuw et al., 2016a; Kim et al., 2020). However, as WM is only one component of EF, our study suggests that future research should focus on other components, such as CF, to enrich existing knowledge.

However, some limitations should be considered when interpreting our findings. First, inhibition is another foundational EF component and reflects the ability to withhold a well-learned prepotent response to enact a subdominant response (Diamond, 2013). In addition to WM and CF, inhibition may play an important role in children's online process, as readers sometimes need to inhibit attention to irrelevant information, such as insignificant background knowledge, for enhanced comprehension to occur. Future studies should focus on such inhibitions to further expand research in this area. Second, and more critically, our study did not account for reading-specific skills, such as word reading and reading comprehension. As noted in the introduction, reading-specific skills are also significant ID factors that influence the reading process (Kuperman et al., 2018; Veldre & Andrews, 2015). Hence, future research should investigate the roles of both EF and reading-specific skills to better elucidate EF's distinct role in children's sentence processing.

Conclusions

The current study showed how student- and text-related characteristics were associated with eye movement and how IDs in children's EF influenced semantic processing during reading comprehension. First, we identified a notable incongruency effect in eye-movement measures, highlighting children's ability to perceive incongruent information. However, children's WM and CF regulate this process. High-WM children were able to more effectively integrate incongruent information, especially when the information in the text was closely related. Children with high CF demonstrate a rapid capacity for discerning incongruent information. Our findings provide preliminary evidence indicating a correlation between EF and variations in sentence processing among students.

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Conflict of Interest Statement

The authors declare that they have no conflict of interest.

Data Availability Statement

Study materials and data are available online at https://osf.io/jzmpv/.

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Appendix A. Summary of Random-Effects Structures in Linear Mixed-Effects Models

Measure	Random-effects structure
First fixation duration	(1 + age + intelligence + CF:association + CF:congruency: association | subject) + (1 + WM:congruency + WM:association + CF: congruency + CF:congruency:association | item)
Gaze duration	(1 + intelligence + WM + CF + WM:association + congruency:association + CF: association | subject) + (WM + WM:congruency + WM:association + congruency: association + CF:congruency + CF:association + CF:congruency:association | item)
Second reading time	(1 + association + WM:association + congruency:association + WM:congruency: association + CF:congruency:association | subject) + (1 + WM: congruency + congruency:association + CF:congruency + WM:congruency: association + CF:congruency:association | item)
Regression probability	(1 + congruency + congruency:association + CF:congruency: association | subject) + (1 + age + intelligence + WM + association + congruency: association | item)

Note: Value 1 represents the inclusion of a random intercept. Abbreviations: CF, cognitive flexibility; WM, working memory.

Nannan Cui is a PhD student in the School of Psychology at Northeast Normal University, Changchun, China.

Yang Wang is a lecturer in the School of Psychology at Xinjiang Normal University, Ürümqi, China. She is also a PhD student in the School of Psychology at Northeast Normal University, Changchun, China.

Jiefei Luo is a PhD student in the School of Psychology at Northeast Normal University, Changchun, China.

Yan Wu is an associate professor in the School of Psychology at Northeast Normal University, Changchun, China.

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Address for correspondence: Yan Wu, School of Psychology, Northeast Normal University, No. 5268, Renmin Street, Changchun, Jilin Province, China. Email: wuy399@nenu.edu.cn

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